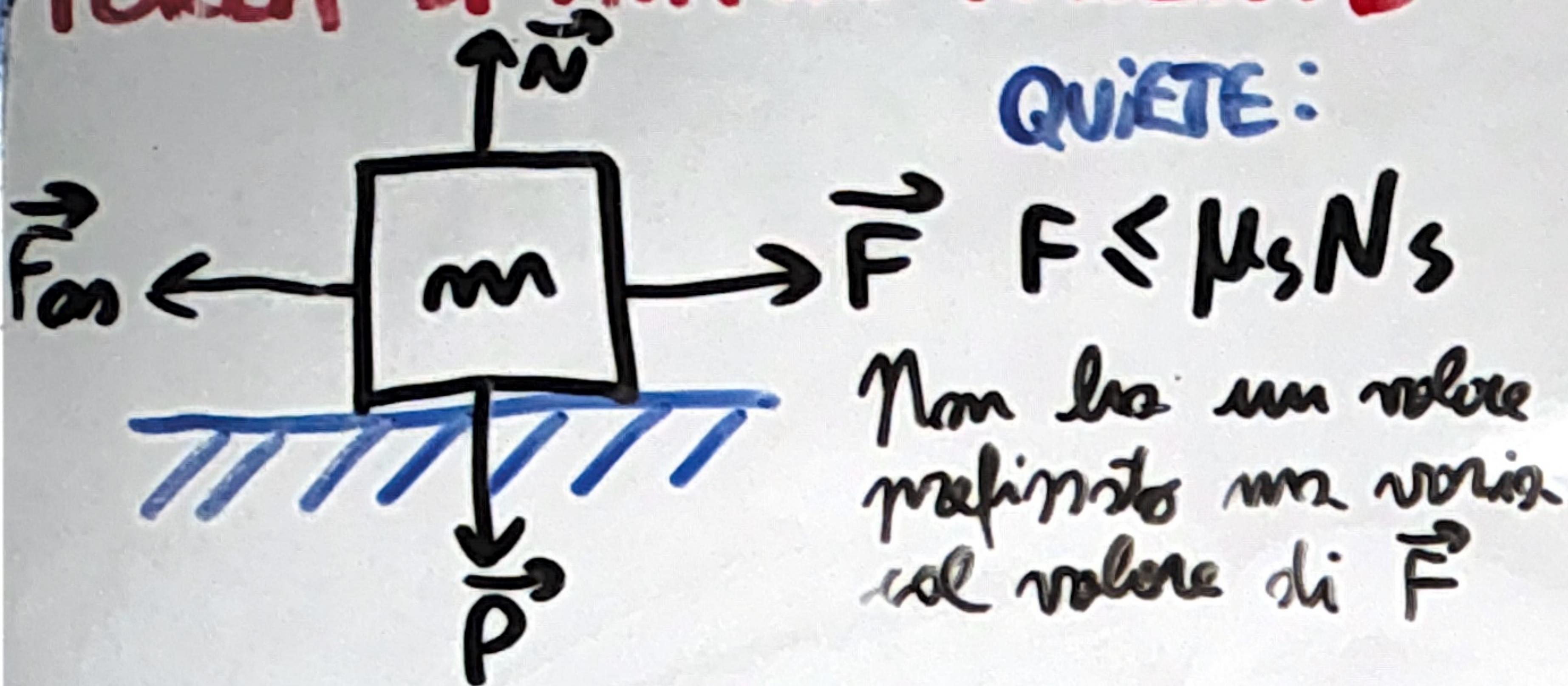


FORZA DI ATTRITO RADENTE



QUIETE:

$$F \leq \mu_s N_s$$

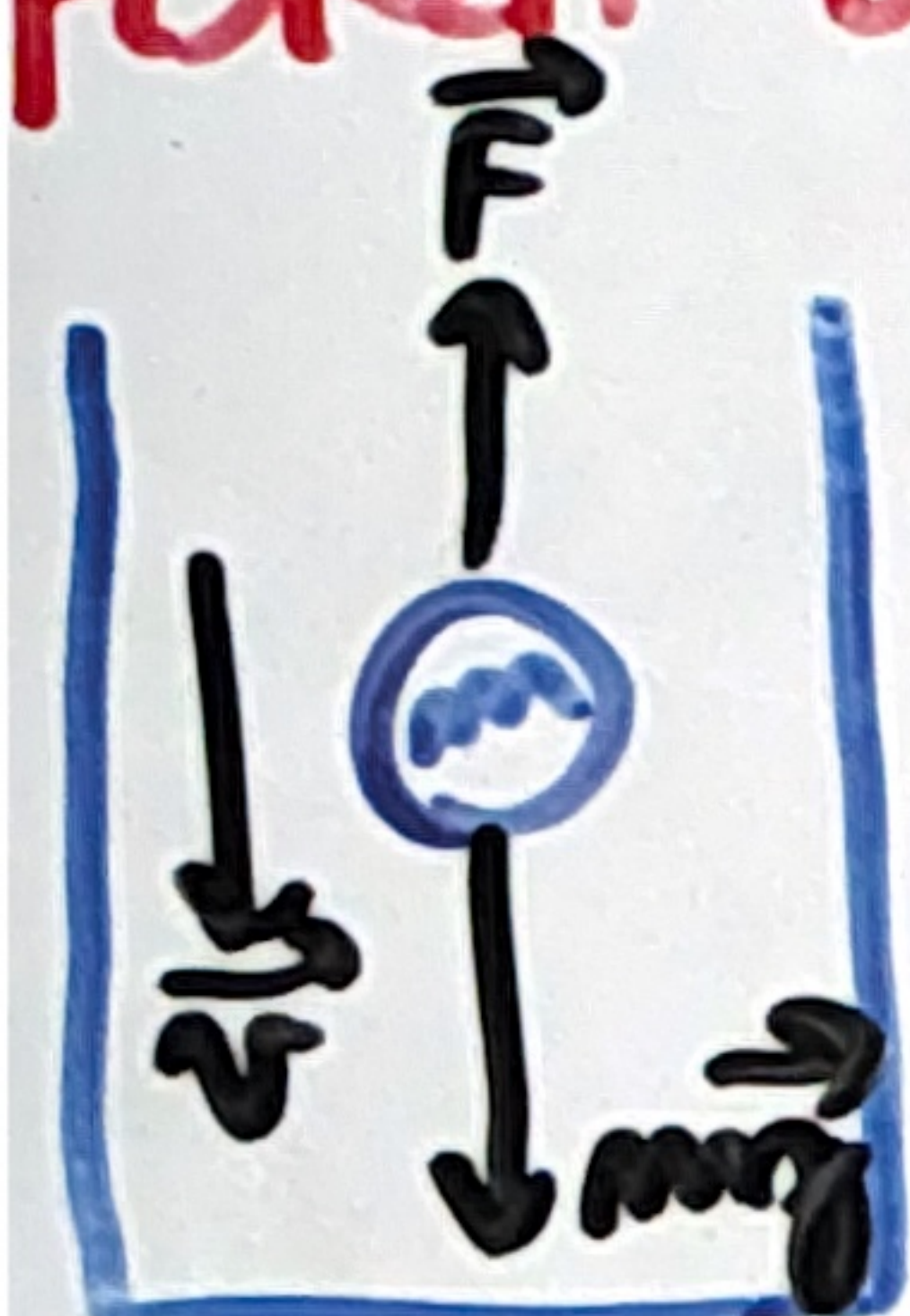
Non ha un valore prefissato ma varia col valore di F

FORZA DI ATTRITO DINAMICA

Quando si supera il valore $\mu_s N$:

$$F - \mu_d N = ma, \text{ con } \mu_d < \mu_s$$

FORZA DI ATTRITO VISCOZO

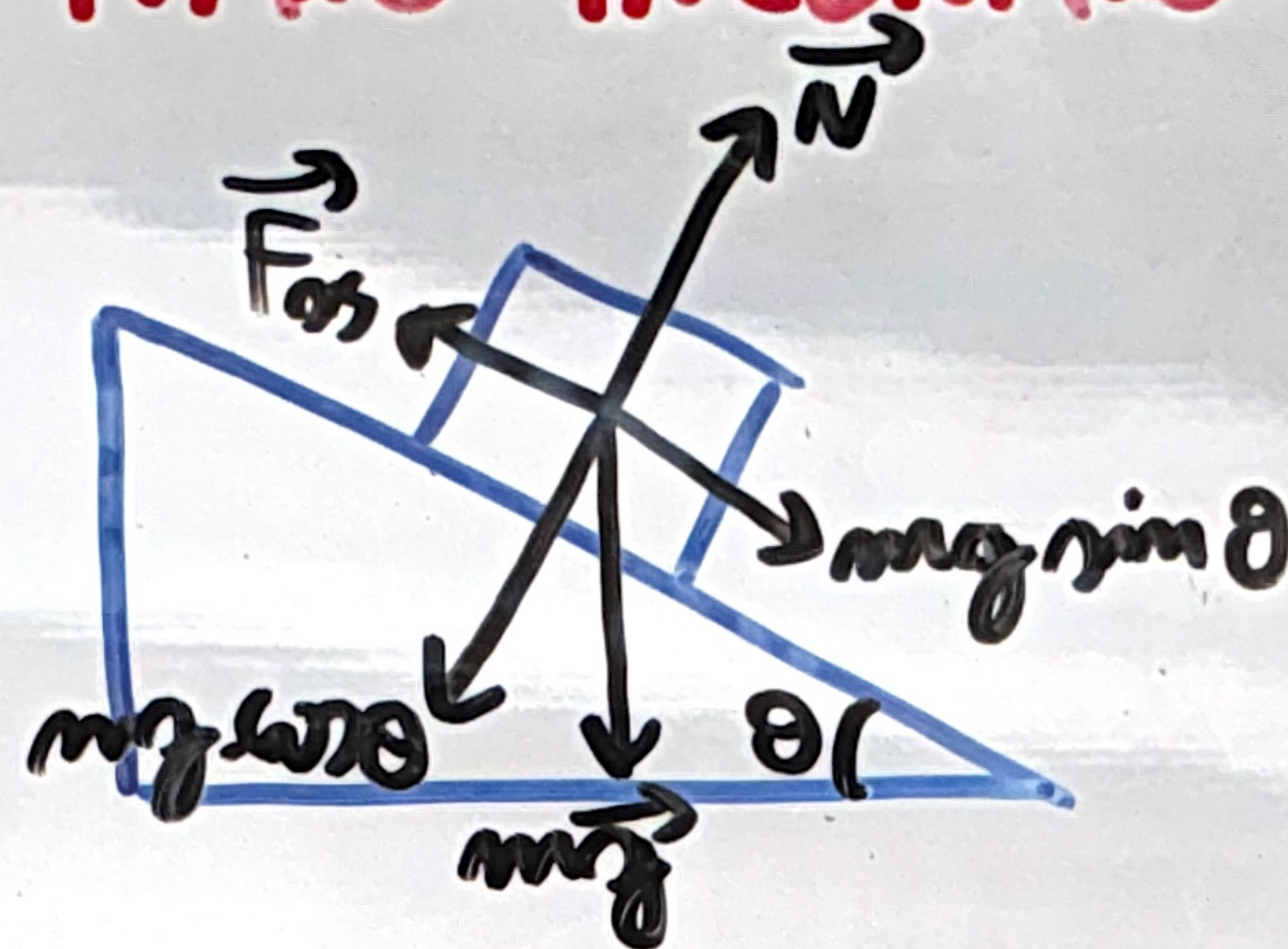


$$v(t) = v_0 e^{-t/\tau}$$

$$\text{con } \tau = \frac{m}{b}$$

$$v(t) = g\tau(1 - e^{-t/\tau})$$

PIANO INCLINATO



ASSENZA DI ATTRITO

$$-mg \cos \theta + N = 0$$

$$mg \sin \theta = ma$$

ATTRITO RADENTE

$$mg \sin \theta \leq F_{os, \max} = \mu_s N = \mu_s mg \cos \theta$$

- equilibrio statico: $\tan \theta \leq \mu_s$

$$-mg \cos \theta + N = 0, \quad mg \sin \theta - \mu_d N = ma$$

$$\rightarrow a = (\sin \theta - \mu_d \cos \theta) g$$

FORZA ELASTICA

$$\vec{F} = -kx \hat{u}_x \quad a = \frac{F}{m} = -\frac{k}{m}x = -\omega^2 x$$

Il sistema costituisce un moto armonico semplice

$$\omega = \sqrt{\frac{k}{m}} \quad T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{m}{k}}$$

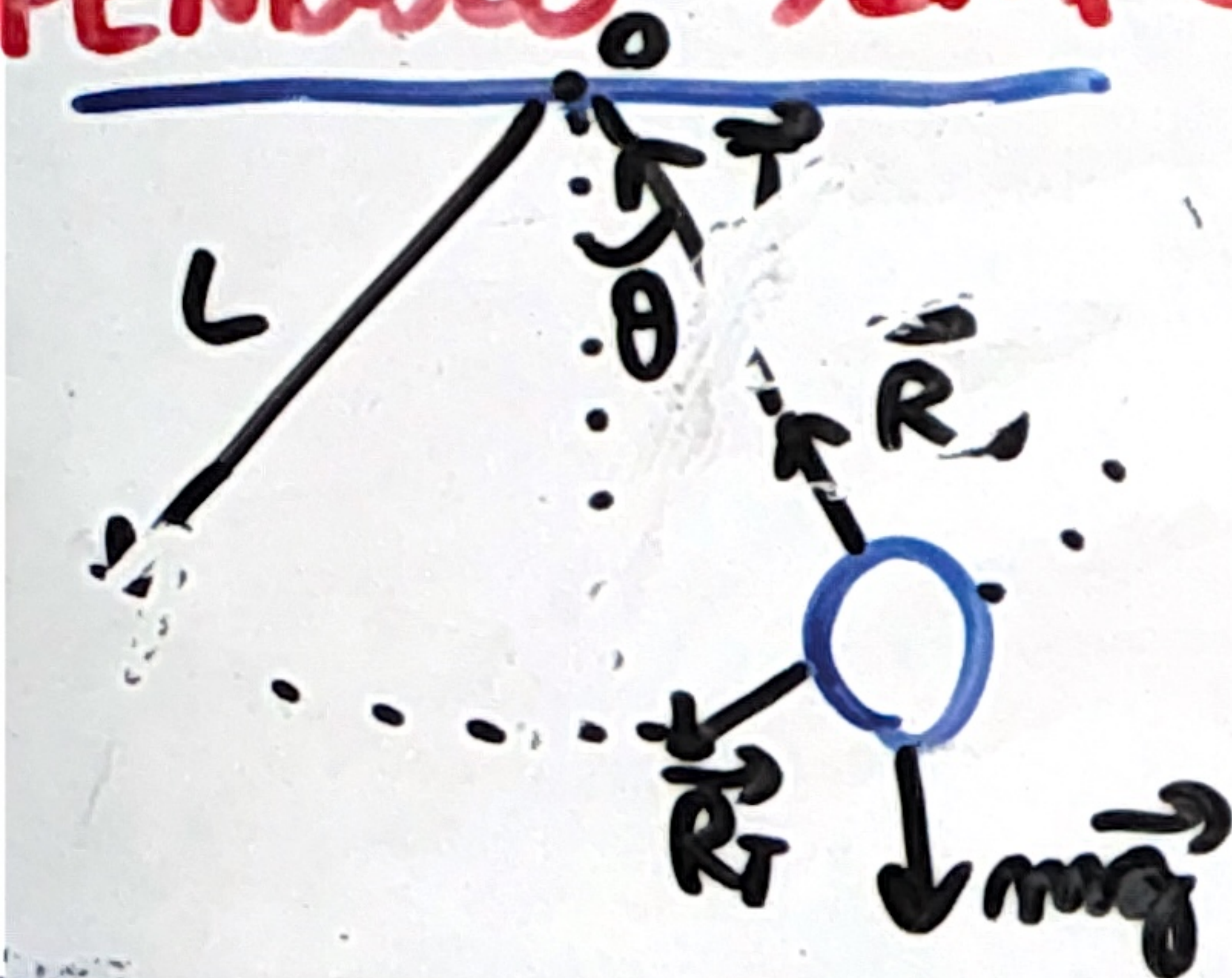
MOLLA ESTESA:

TEOR. DELL'IMPULSO

$$F_x \Delta t = m(v - v_0)$$

$$\text{SE } v = \omega x \Rightarrow F_{tot} = 0$$

PENDOLO SEMPLICE



$$R_T = -mg \sin \theta = ma_T$$

$$a_T = L\alpha$$

$$R_N = T_F - mg \cos \theta = ma_N$$

$$a_N = \frac{v^2}{L}$$

$$T_F = m \left[g \cos \theta(t) + \frac{v^2(t)}{L} \right]$$

$$\omega = \omega_0 \cos(\omega t + \phi) \quad v = L \omega_0 \sin(\omega t + \phi)$$

PER PICCOLE OSCILLAZIONI

$$\theta = \theta_0 \sin(\omega t + \phi)$$

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{L}{g}}$$

$$a = L\theta = L\theta_0 \sin(\omega t + \phi)$$